

Project Name: Kumasi Kingdom City
Duration: 1st January – 1st April, 2025
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3D Assets Production and Optimization – Kumasi Kingdom City

Date: 01/01/2025 – 24/08/2025

Buildings

- Optimized **ten (10) ordinary buildings** in the Kumasi Kingdom City for use in VR.
- Optimized **three (3) royal-style structures** in the Kumasi Kingdom City for use in VR.
- Optimized **a storey building** in the Kumasi Kingdom City for use in VR.
- Optimized **an Apokoo building** in the Kumasi Kingdom City for use in VR.

Denkyirahene's Courtyard

- Set up scene of the Denkyirahene's house in Unreal Engine complete with fruits, vegetables, meat and fish.
- Modeled and textured one (1) calabash with gold nuggets.
- Modeled and textured four (4) house types .
- Modeled and textured one (1) large traditional drum.
- Modeled and textured one (1) small traditional drum.
- Modeled and textured two (2) clay pots.
- Textured two (2) crates.
- Modeled and textured two (2) bows and one (1) arrow
- Modeled and textured one (1) quiver
- Modeled and textured one (1) sword with a sheath
- Modeled and textured one (1) cutlass
- Modeled and textured one (1) spear
- Modeled and textured a set of ten (10) traditional drums

Osei Tutu's House

- Set up scene of Osei Tutu's house in Unreal Engine, complete with traditional artifacts and his weapons.
- Modeled and textured the house.
- Modeled and textured Osei Tutu's bed.
- Modeled and textured a box full of clothes and jewelry.
- Modeled and textured Osei Tutu's chair.
- Modeled and textured two (2) clay pots with scrolls.
- Modeled and textured Okomfo Anokye's staff.
- Modeled and textured one (1) lantern.
- Modeled and textured one (1) table.
- Modeled and textured a map on the table.
- Modeled and textured Osei Tutu's drinking flask.
- Modeled and textured Osei Tutu's hunting bag.
- Cleaned up **two (2) artifacts** obtained from 3D scans.

Golden Stool Scene

- Set up the scene of the Golden stool in Unreal Engine.
- Created a detailed textured model of the **Golden Stool** of the Ashanti Kingdom with the symbolic **black bell**.
- Modeled and textured **three (3) royal chairs** using reference images and artistic inspiration. Two will be for Osei Tutu and the Denkyirahene, while the other will be used by the other chiefs.
- Modeled and textured **Okomfo Anokye's whisk** using reference images and artistic inspiration.
- Modeled and textured two (2) buildings.
- Modeled and textured **three (3) royal swords** using reference images and artistic inspiration.
- Modeled and textured **eleven (11) royal umbrellas** using reference images and artistic inspiration for the various chiefs of the Ashanti Kingdom.
- Modeled and textured **seven (7) staffs** for the staff bearers using reference images and artistic inspiration.
- Cleaned up **two (2) traditional drums** obtained from 3D scans.
- Textured **five (5) traditional black stools**.
- Cleaned up and reconstructed **ten (10) traditional stools** obtained from 3D scans.
- Modeled and textured a sacrificial stone.

Marketplace Scene

- Set up the Marketplace scene in Unreal Engine, complete with sellers' stands filled with food and other items for sale.
- Modeled and textured one (1) barrel.
- Modeled and textured one (1) box.
- Modeled and textured a piece of cloth.
- Modeled and textured a market building.
- Modeled and textured three (3) mats
- Modeled and textured a wooden stand
- Modeled and textured a structured made from wood and cloth
- Modeled and textured one (1) bench
- Modeled and textured five (5) basket types
- Modeled and textured five (5) bowl types
- Modeled and textured a cup
- Modeled and textured a gourd
- Modeled and textured a jug
- Modeled and textured a knife
- Modeled and textured three (3) plate types
- Modeled and textured one (1) spoon
- Modeled and textured a stack of cloth
- Modeled and textured a vase

Shrine Scene

- Set up the Shrine scene in Unreal Engine.
- Modeled and textured a slain dog.
- Modeled and textured a basket full of bones and a skull.
- Modeled and textured the shrine building.
- Modeled and textured twenty (20) totems.

Challenge Faced

Balancing Detail and Performance: Maintaining a reasonable polycount and texture resolution for VR implementation while still being able to portray the rich intricate details of the Ashanti traditional items.

OSEI TUTU'S HOUSE







DENKYIRAHENE'S COURTYARD









GOLDEN STOOL SCENE







MARKETPLACE SCENE











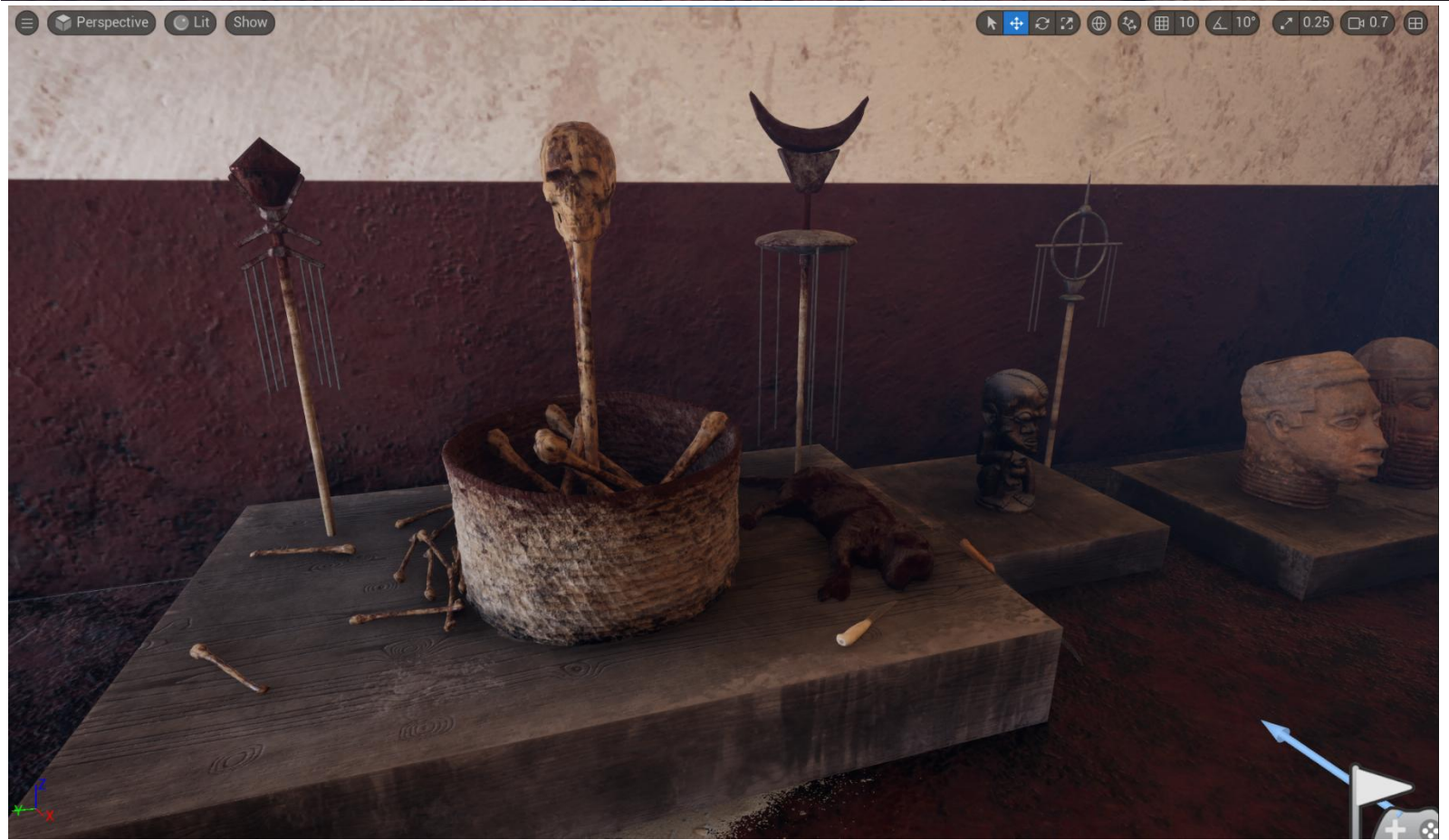








SHRINE SCENE



Shift+F1 for Mouse Cursor







THE GOLDEN STOOL



OKOMFO ANOKYE'S WHISK



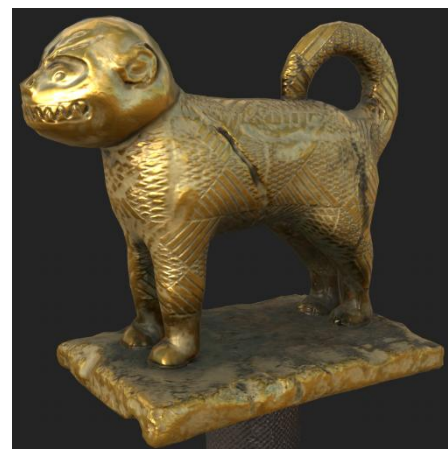
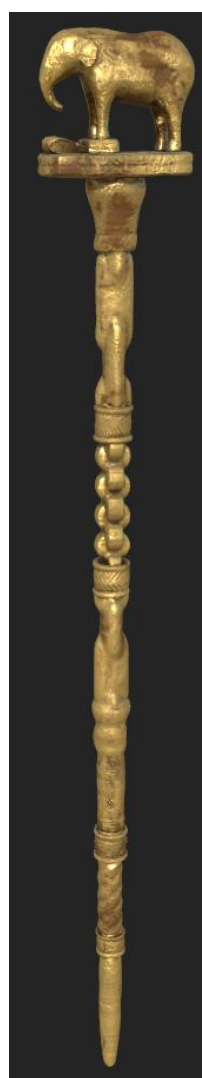
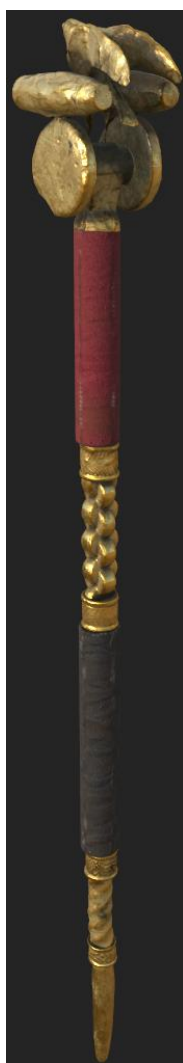
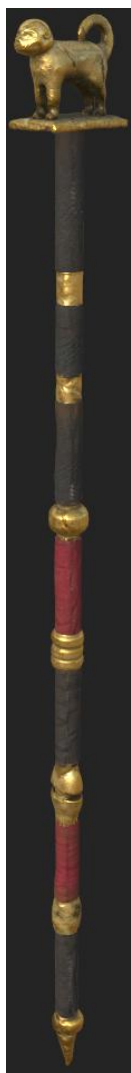
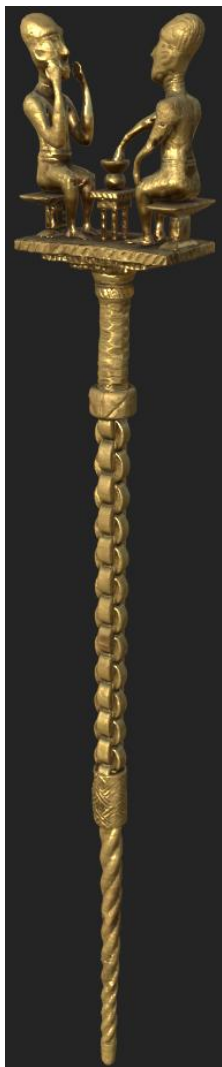
ROYAL SWORDS



ROYAL UMBRELLAS



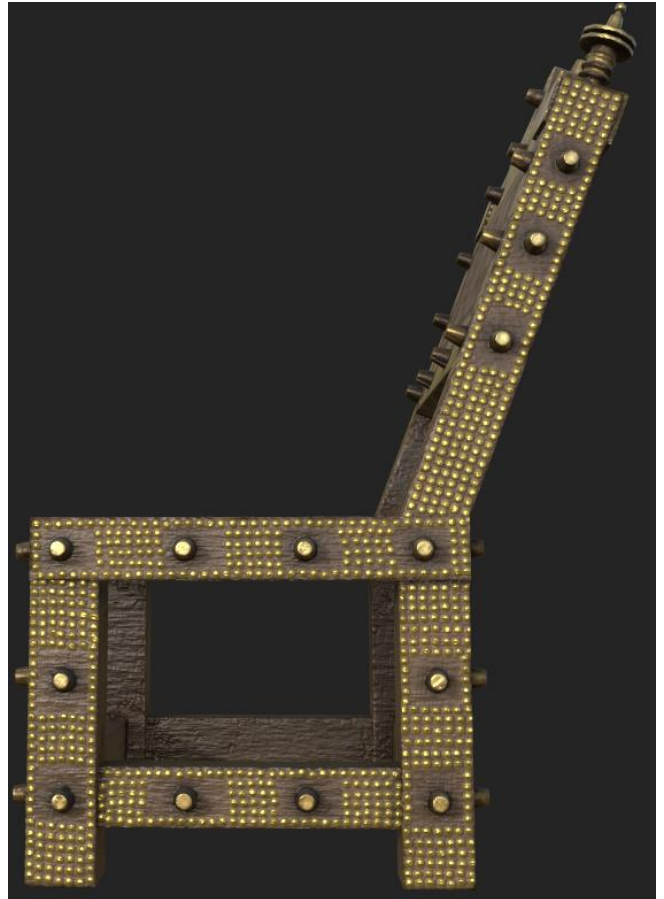
SOME ROYAL STAFFS



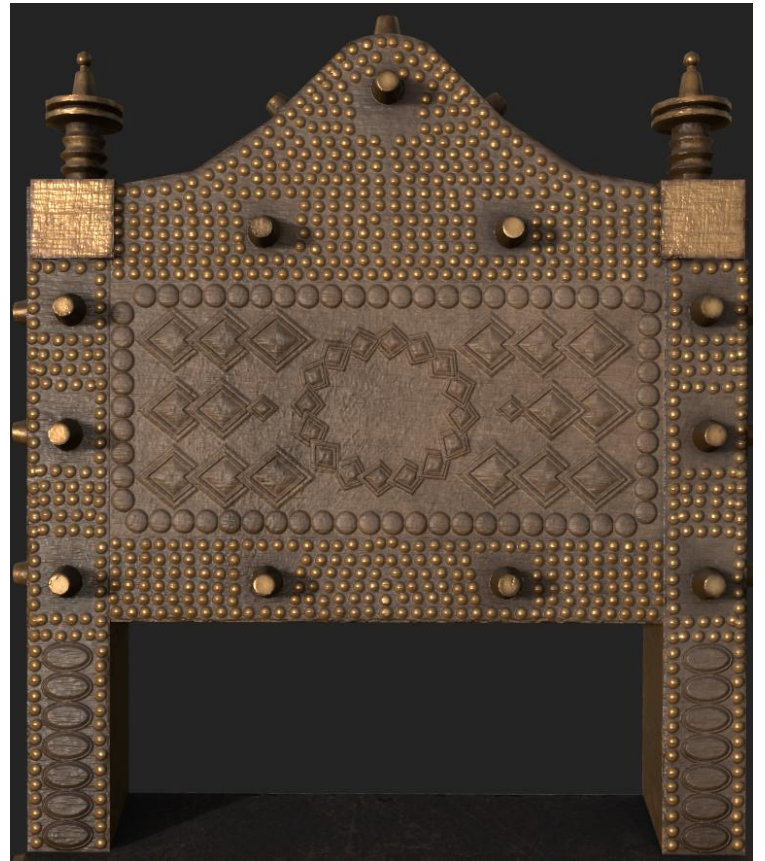
OSEI TUTU'S CHAIR



DENKYIRAHENE'S CHAIR



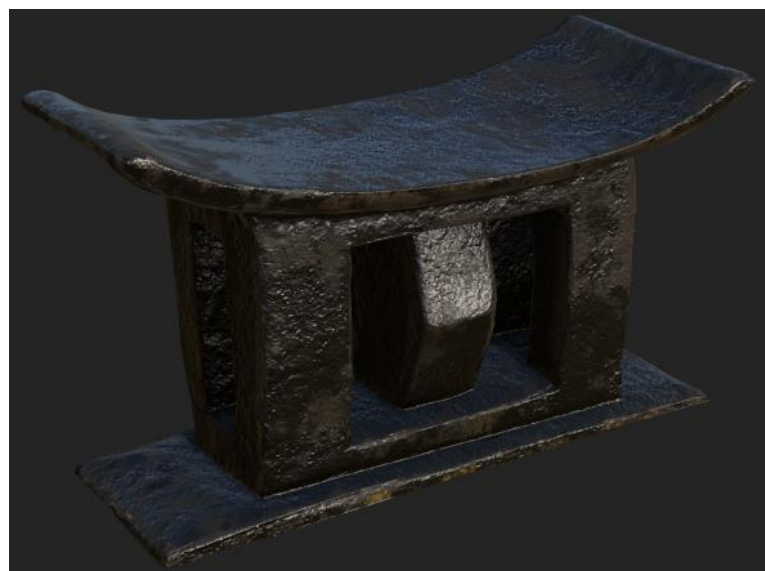
CHAIR FOR OTHER CHIEFS



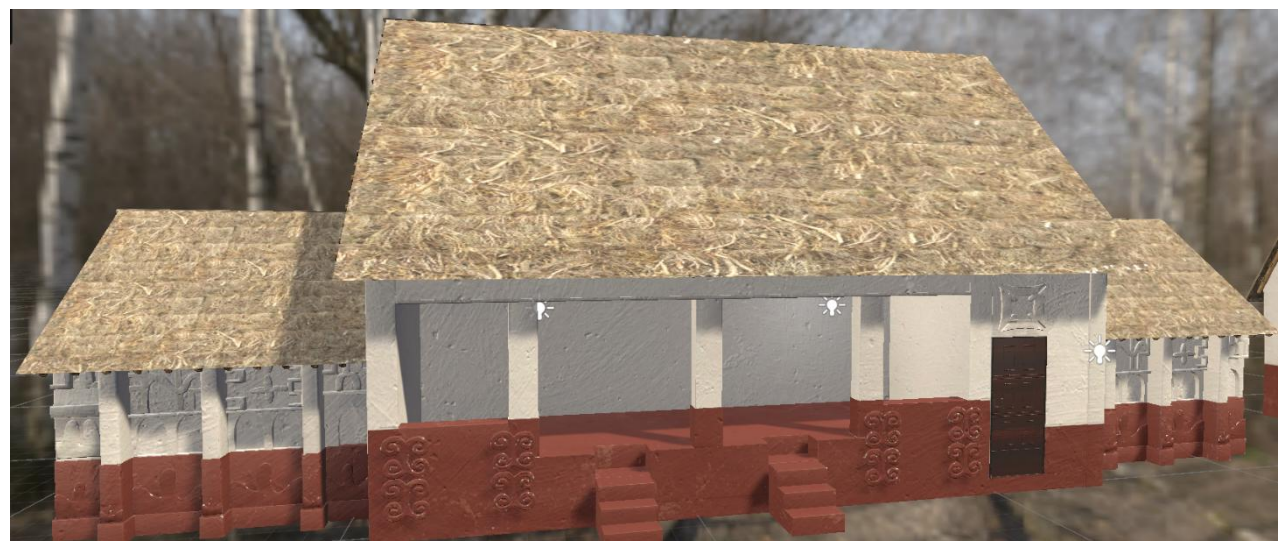
TRADITIONAL STOOLS

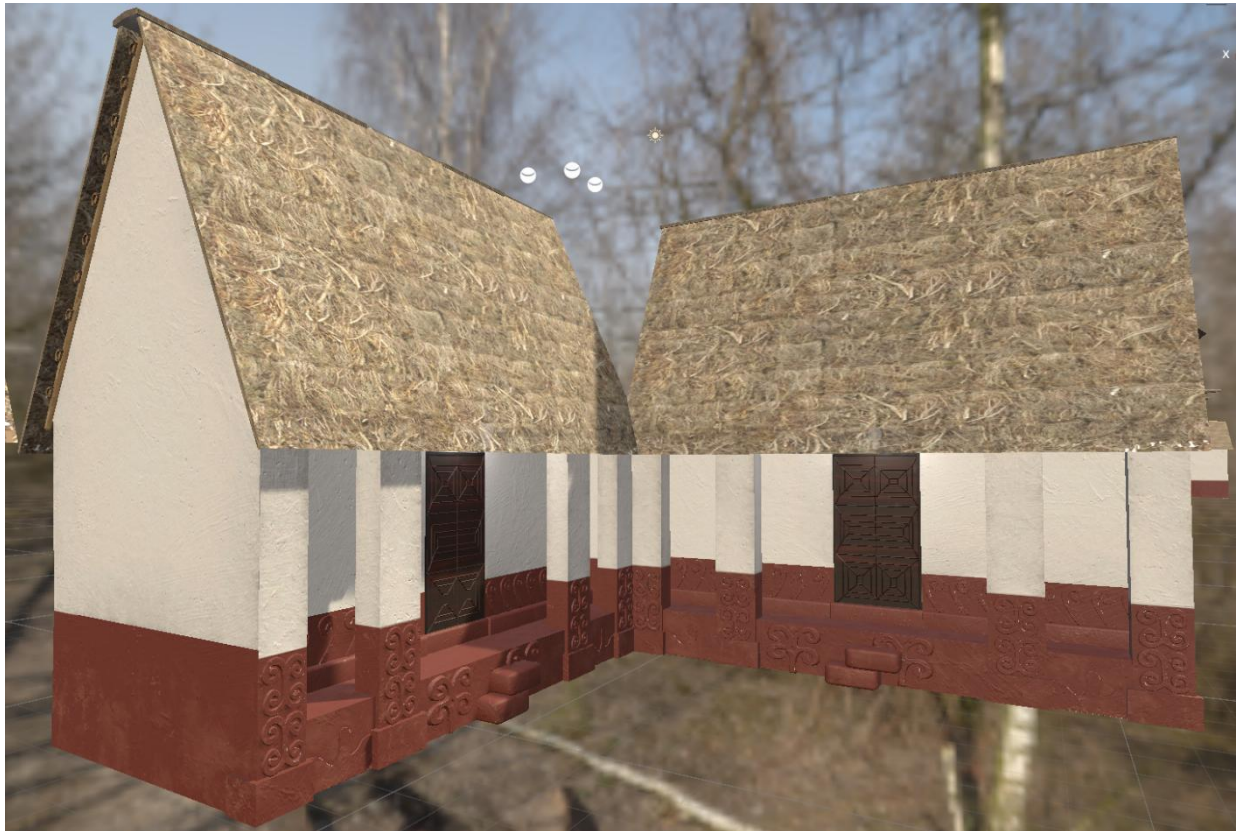
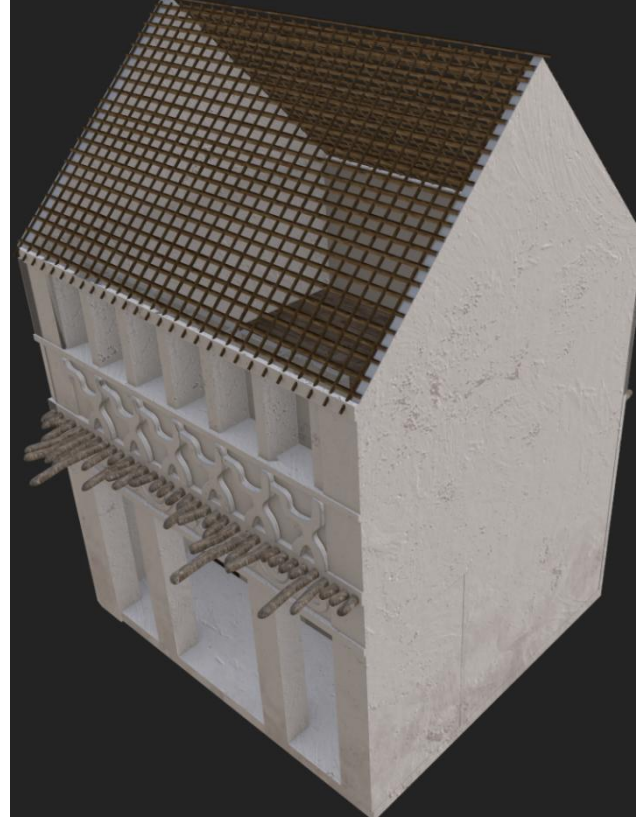


BLACK STOOLS



SOME KUMASI KINGDOM CITY BUILDINGS





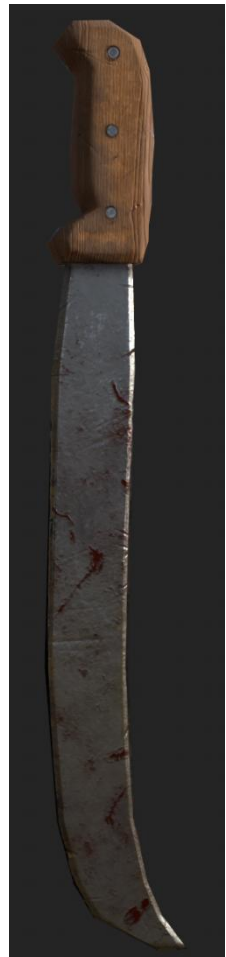
SET OF TEN (10) TRADITIONAL DRUMS

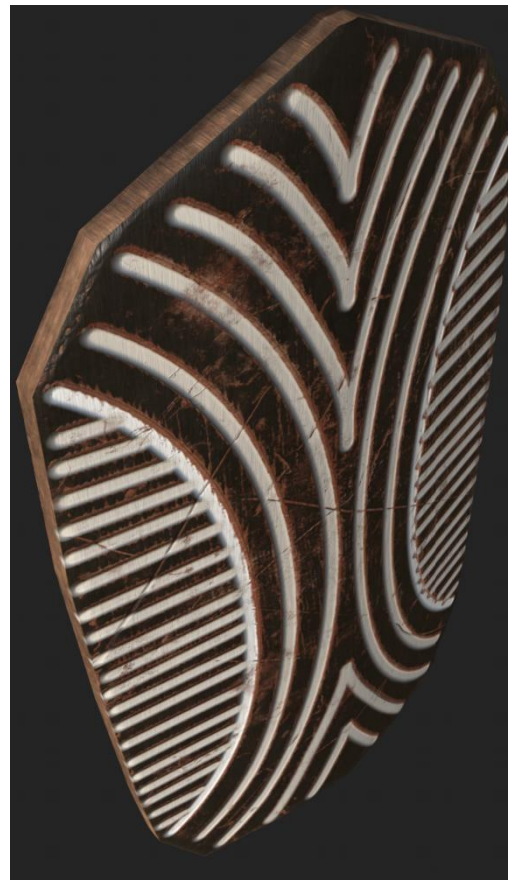




SET OF ASHANTI WEAPONS







LARGE DRUM

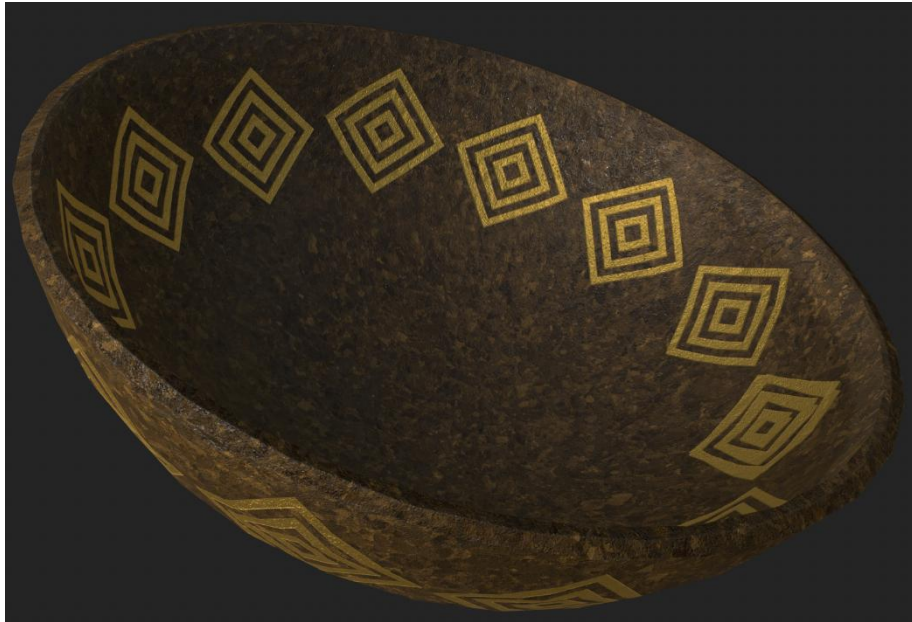




CLAY POTS



CALABASH



SACRIFICIAL STONE

